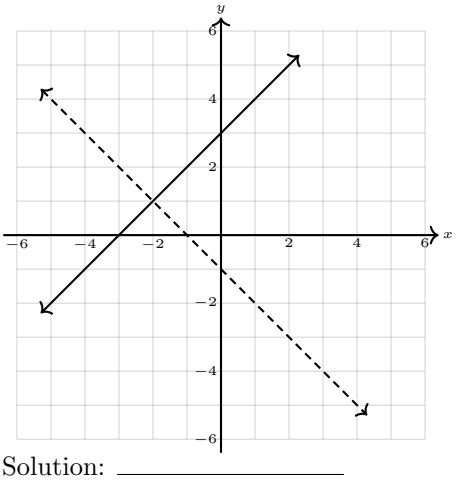
Math 15 — Unit 8: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

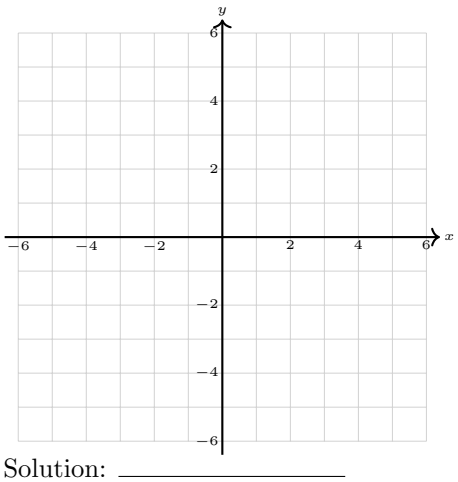
Part A — Reading and Solving by Graphing

1. The system $y = x+3$ and $y = -x-1$ is graphed below. State the solution as an ordered pair.



Solution: _____

2. Consider the system $y = x+2$ and $y = -x+4$. Graph both equations without using technology, and state the solution to the system.



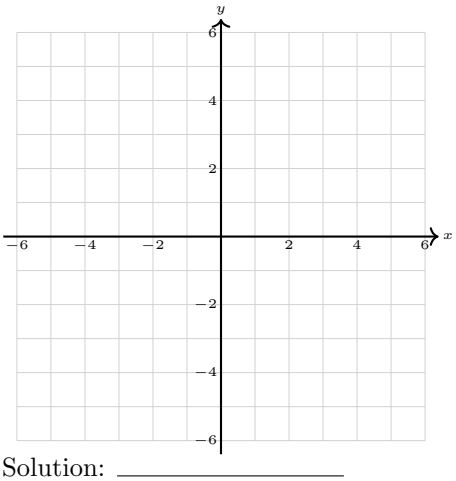
Solution: _____

3. Consider the system $x - y = 3$ and $y = -x + 5$.

(a) Rewrite $x - y = 3$ in slope-intercept form:

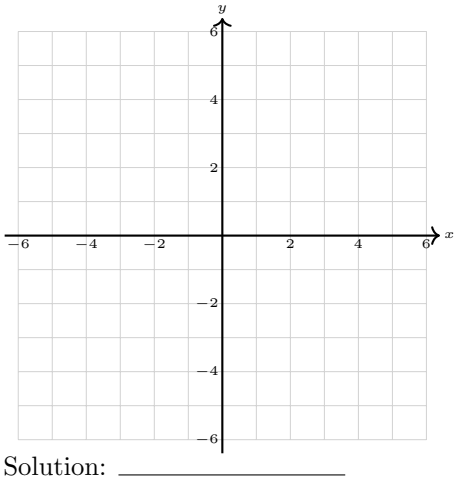
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(b) Graph both equations and state the solution.

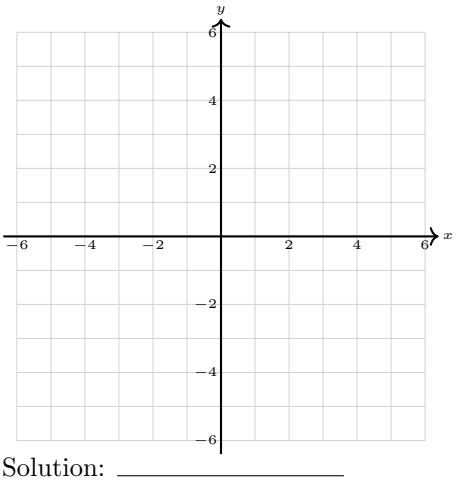


Solution: _____

4. Consider the system $y = -x+2$ and $y = 3x-6$. Graph both equations without using technology, and state the solution to the system.



Solution: _____



5. Consider the system $x + y = 5$ and $x - y = 1$.
(a) Rewrite each equation in slope-intercept form.

(b) Graph both equations on the grid at left and state the solution to the system.

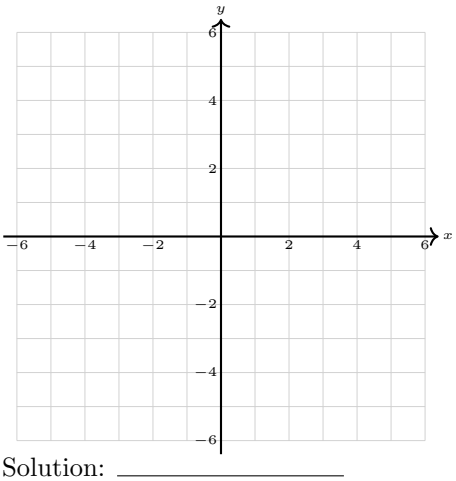
Solution: _____

Part B — Checking a Proposed Solution

6–8. Determine whether the given ordered pair is the solution to the system. Show the substitution into both equations.
6. $(4, 2)$ for $y = 2x - 6$ and $y = -x + 6$. 7. $(1, 5)$ for $y = 3x + 1$ and $y = -2x + 9$. 8. $(3, -1)$ for $2x + y = 5$ and $y = x - 4$.

9. In your own words, explain why it is important to check a proposed solution in **both** original equations of a system, rather than just one.

Part C — Rearranging, Graphing and Checking



10. Consider the system $4x - y = 3$ and $y = -x + 2$.
(a) Rewrite $4x - y = 3$ in slope-intercept form.

(b) Graph both equations on the grid at left and state the solution.
(c) Check the solution in both **original** equations.

Solution: _____

Part D — Multiple Choice and Numerical Response

11. *(Multiple Choice)* The ordered pair (x, y) which satisfies the system $y = 2x - 3$, $y = -x + 9$ is
A. $(4, 5)$ B. $(5, 4)$ C. $(-4, 5)$ D. $(4, -5)$

12. *(Multiple Choice)* On the graph of a system of two linear equations, the solution to the system is
A. the y-intercept of either line B. the x-intercept of either line
C. the point where the two lines cross D. the slope of either line

13. *(Numerical Response)* The system $y = 5x - 8$ and $y = -2x + 13$ has solution (x, y). (Make a table of values for both equations and find where the y-values match.) The value of $x + y$ is _____.

14. *(Numerical Response)* A system's two equations are $y = -4x + 9$ and $y = x - 1$. Using a table of values, the value of y at the solution is _____.

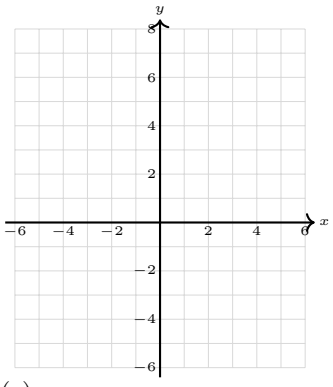
Answer Key. 1. $(-2, 1)$. **2.** $(1, 3)$. **3.** (a) $y = x - 3$ (b) $(4, 1)$. **4.** $(2, 0)$. **5.** (a) $y = -x + 5$, $y = x - 1$ (b) $(3, 2)$. **6.** Yes — $2(4) - 6 = 2$ ✓ and $-4 + 6 = 2$ ✓. **7.** No — $3(1) + 1 = 4 \neq 5$, so the first equation already fails. **8.** Yes — $2(3) + (-1) = 5$ ✓ and $3 - 4 = -1$ ✓. **9.** A point can satisfy one equation without lying on the other line, so checking a single equation cannot confirm it solves the *system*. **10.** (a) $y = 4x - 3$ (b) $(1, 1)$ (c) $4(1) - 1 = 3$ ✓, $-1 + 2 = 1$ ✓. **11.** A ($(4, 5)$ checks in both: $2(4) - 3 = 5$, $-4 + 9 = 5$). **12.** C. **13.** 10 (solution $(3, 7)$). **14.** 1 (solution $(2, 1)$). *Lesson examples:* **Ex1** $(1, 4)$; **Ex2** $(2, -1)$; **Ex3** $y = -2x + 1$, $(2, -3)$; **Ex4** (a) yes (b) no; **Ex5** $y = -x + 3$, $(2, 1)$.

Lesson 2 Assignment *Show your work.*

Part A — Graph and Classify

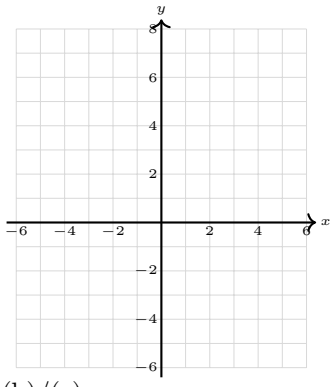
1–2. For each system: (a) state the slope of each line; (b) graph both lines on the grid; (c) state the number of solutions — for Q1 also give the solution as an ordered pair, and for Q2 explain why the lines never cross.

1. $y = x + 1,\ \ y = -2x + 7$ (a) m_1____ m_2____



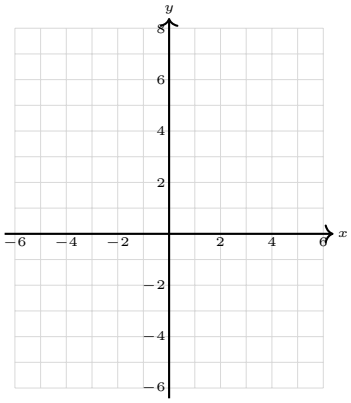
(c)

2. $y = -x + 3,\ \ y = -x - 2$ (a) m_1____ m_2____



(b)/(c)

3. Consider the system $x - y = 2$ and $2x - 2y = 4$.



(a) Rewrite both equations in slope y-intercept form.

(b) Graph both lines on the grid at left.

(c) State the number of solutions.

Part B — Classifying Without Graphing

4. Without graphing, state whether each system has one solution, no solution, or infinitely many solutions.

(a) $y = -3x + 7$
$y = -3x + 1$

(b) $y = 6x - 5$
$y = -2x + 11$

(c) $y = -5x + 3$
$y = -5x + 3$

(d) $y = 4$
$y = -2$

5. Rearrange each system into slope y-intercept form, then state whether it has one solution, no solution, or infinitely many solutions.

(a) $3x + y = 5,\ \ y = -3x + 9$

(b) $x - y = 6,\ \ 2x - 2y = 12$

(c) $4x - y = 1,\ \ x + y = 9$

Part C — Explain and Build

6. How can you tell, just by looking at the *graph* of a system of two linear equations, whether the system has one solution, no solution, or infinitely many solutions?

7. How can you tell, by writing a system in the form $y = mx + b$, whether the system has one solution, no solution, or infinitely many solutions?

8. The equation $y = 2x - 5$ is given. Write a second equation that pairs with it to form a system with:
(a) no solution (b) one solution (c) infinitely many solutions

Part D — Multiple Choice and Numerical Response

9. *(MC)* The system $y = -4x + 7$, $y = -4x - 2$ has
A. no solution B. one solution
C. infinitely many solutions D. two solutions

10. *(NR)* The value of k, $k \in N$, for which the system $3x + ky = 9$ and $6x + 4y = 18$ has infinitely many solutions, is _____.

11. *(NR)* The value of a, $a \in N$, for which the system $ax + 2y = 10$ and $3x + y = 4$ has no solution, is _____.

Part E — Applications and Reasoning

12. Keisha says the system $y = 5x - 2$ and $10x - 2y = 4$ has no solution because the two equations look different. Is she correct? Explain, showing your work.

13. Texting Plan A costs $y = 0.10x + 5$ per month (where x is the number of texts sent), and Plan B costs $y = 0.10x + 8$ per month. Without solving algebraically, determine whether there is a number of texts for which the two plans cost the same. Explain using slopes and intercepts.

14. A downtown parking garage charges $y = 3x+4$ (where x is the number of hours parked). A nearby lot charges $y = 5x$. By graphing both lines (or with a table of values), determine whether there is a number of hours for which the two costs are the same, and if so, find it.

15. Determine the value of c for which the system $4x - 2y = c$ and $y = 2x - 3$ has infinitely many solutions.

Answer Key. 1. (a) 1 and -2 (b) lines cross once (c) one solution, $(2, 3)$. **2.** (a) -1 and -1 (b) same slope, different intercepts, so parallel (c) no solution. **3.** (a) $y = x - 2$ and $y = x - 2$ (b) identical lines (c) infinitely many solutions. **4.** (a) no solution (b) one solution (c) infinitely many solutions (d) no solution (two horizontal lines). **5.** (a) $y = -3x + 5$, $y = -3x + 9$: no solution (b) $y = x - 6$, $y = x - 6$: infinitely many (c) $y = 4x - 1$, $y = -x + 9$: one solution. **6.** Cross at one point $\to$ one solution; parallel (never touch) $\to$ no solution; one line on top of the other $\to$ infinitely many. **7.** Compare slopes first: different $\to$ one solution. Equal slopes: compare intercepts — different $\to$ no solution, equal $\to$ infinitely many. **8.** (a) e.g. $y = 2x - 9$ (b) e.g. $y = x - 5$ (c) e.g. $4x - 2y = 10$. **9.** A. **10.** 2. **11.** 6. **12.** No; $10x - 2y = 4$ simplifies to $y = 5x - 2$, the same line, so the system has infinitely many solutions. **13.** No; both have slope 0.10 but intercepts 5 and 8, so the lines are parallel and the costs are never equal. **14.** Slopes 3 and 5 differ, so one solution: the lines cross at $(2, 10)$ — equal at 2 h, \$10. **15.** $c = 6$.

Lesson 3 Assignment *Show your work.*

Part A — Substitute, Solve, Back-Substitute

1. Solve each system by substitution. Check your solution in both equations.

(a) $y = x + 3$
$2x + y = 12$

(b) $y = x - 5$
$3x + 2y = 20$

(c) $x = y + 4$
$2x - y = 11$

(d) $x = 2y + 1$
$3x - y = 13$

2. Solve each system by substitution. Check your solution in both equations. (One answer is negative — that is normal.)

(a) $y = x + 7$
$3x - y = 1$

(b) $y = 3x + 2$
$x + y = 14$

(c) $x = y - 6$
$x + y = 4$

(d) $x = 4y + 3$
$2x - y = 20$

3. Solve each system by substitution. Check your solution in both equations.

(a) $y = 2x - 3$
$4x - y = 9$

(b) $y = -x + 8$
$5x + 2y = 19$

(c) $x = 3y - 5$
$2x + y = 4$

Part B — Checking, Multiple Choice, Numerical Response

4. *(Checking Practice)* A student's claimed solution is given. Substitute it into both original equations to test it. If it fails either one, find the correct solution by substitution.

(a) System: $y = x + 6$, $3x - y = 2$.
Claimed solution: $(4, 9)$.

(b) System: $x = 2y - 1$, $x + y = 11$.
Claimed solution: $(7, 4)$.

5. *(Multiple Choice)* If $x = y + 3$ and $2x - y = 11$, then y is equal to

A. 5 B. 2 C. 8 D. -5

6. *(Numerical Response)* If $y = 4x - 1$ and $3x + y = 20$, then the value of $x + y$ is _____.

Answer Key. 1. (a) $(3, 6)$ (b) $(6, 1)$ (c) $(7, 3)$ (d) $(5, 2)$. **2.** (a) $(4, 11)$ (b) $(3, 11)$ (c) $(-1, 5)$ (d) $(11, 2)$. **3.** (a) $(3, 3)$ (b) $(1, 7)$ (c) $(1, 2)$. **4.** (a) $(4, 9)$ is incorrect — it fails both equations; correct solution $(4, 10)$ (b) $(7, 4)$ is correct — it satisfies both equations. **5.** A. **6.** 14 (from $x = 3$, $y = 11$).

Lesson 4 Assignment *Show your work.*

Part A — Choosing a Variable and Solving

1. State which variable you would isolate, and in which equation, to avoid fractions. Do not solve.

(a) $3x + y = 7, \quad 2x - 5y = 4$

(b) $x - 4y = 9, \quad 3x + 2y = 5$

(c) $5x + 2y = 11, \quad x + 3y = -2$

(d) $2x - y = 6, \quad 4x + 3y = 1$

2. Solve each system completely by isolating a variable first. Check your solution.

(a) $x + 4y = 15, \quad 3x - 2y = 3$

(b) $2x - y = 9, \quad 3x + 5y = 7$

(c) $x - 5y = 13, \quad 4x + 3y = 6$

(d) $5x + y = 2, \quad 2x - 3y = 11$

3. Solve each system completely. In each one the variable you isolate has a coefficient of -1 — watch your signs.

(a) $3x - y = 11, \quad 2x + 5y = 13$

(b) $5x - y = 7, \quad 3x + 4y = 18$

(c) $2x - y = 7, \quad 4x + 3y = 9$

4. In each system *both* variables have a coefficient of 1 or -1 somewhere, so either choice is reasonable. Solve, then check in *both* original equations.

(a) $x + 2y = 7, \quad 4x - y = 10$

(b) $2x + y = 9, \quad x - 3y = 1$

5. Solve each system completely. No coefficient is 1 or -1, so isolating creates a fraction partway through — carry it carefully and it resolves to a whole number.

(a) $2x + 5y = 16, \quad 3x - 2y = 5$

(b) $4x - 3y = 3, \quad 2x + 5y = 21$

(c) $3x + 4y = 25, \quad 2x - 3y = -6$

Part B — Word Problems

6. The sum of two numbers is 22. Four times the smaller number minus the larger number is 3. Let L be the larger number and S the smaller number.
(a) Write a system of two equations for this situation. (b) Solve the system to find both numbers.

7. At the talent show, adult tickets cost $\$x$ and student tickets cost $\$y$. On Friday, 3 adult and 4 student tickets sold for \$51. On Saturday, 2 adult and 5 student tickets sold for \$48.
(a) Write a system for the two nights' sales. (b) Solve to find the cost of each ticket type.

8. A rectangle has length l (cm) and width w (cm). Twice the length plus three times the width is 66 cm. The length is 3 cm more than the width.
(a) Write a system of two equations for the rectangle. (b) Solve to find the length and the width.

Part C — Multiple Choice, Numerical Response, and Extension

9. *(Multiple Choice)* If $x + 4y = 18$ and $3x - 2y = 12$, then the value of $x - y$ is
A. 9 B. 3 C. -3 D. 15

10. *(Multiple Choice)* When solving a system that includes $5x - y = 12$, a valid isolation of y is
A. $y = 5x - 12$ B. $y = 12 - 5x$ C. $y = 5x + 12$ D. $y = -5x - 12$

11. *(Numerical Response)* If $2x + y = 9$ and $3x - 4y = 8$, then the value of $x + y$ is _____.

12. *(Numerical Response)* If $x - 3y = -2$ and $4x + y = 31$, then the value of $x - y$ is _____.

13. *(Extension)* Solve this system of three equations by substitution: $3p-q+2r=16,\ r-p=5,\ q-2p=3$. *Hint:* rearrange the second and third equations — with no fractions — to give r and q in terms of p, then substitute both into the first.

14. *(Mixed Review)* Solve each system completely, choosing the most efficient variable to isolate.
(a) $x+3y=7,\quad 2x-y=7$ (b) $3x+2y=12,\quad 4x-5y=-7$

Answer Key. 1. (a) y in Eq. 1 (b) x in Eq. 1 (c) x in Eq. 2 (d) y in Eq. 1. **2.** (a) $(3,3)$ (b) $(4,-1)$ (c) $(3,-2)$ (d) $(1,-3)$. **3.** (a) $(4,1)$ (b) $(2,3)$ (c) $(3,-1)$. **4.** (a) $(3,2)$ (b) $(4,1)$. **5.** (a) $(3,2)$ (b) $(3,3)$ (c) $(3,4)$. **6.** (a) $L+S=22$, $4S-L=3$ (b) $L=17$, $S=5$. **7.** (a) $3x+4y=51$, $2x+5y=48$ (b) adult \$9, student \$6. **8.** (a) $2l+3w=66$, $l-w=3$ (b) $l=15$ cm, $w=12$ cm. **9.** B. **10.** A. **11.** $x=4$, $y=1$, so $x+y=5$. **12.** $x=7$, $y=3$, so $x-y=4$. **13.** $p=3$, $q=9$, $r=8$. **14.** (a) $(4,1)$ (b) $(2,3)$.

Lesson 5 Assignment *Show your work.*

Part A — Identify, then Eliminate by Subtraction

1. State whether you would add or subtract to eliminate a variable, and name the variable eliminated. Do not solve.

(a) $4x + 3y = 17, \quad 4x - 5y = 1$

(b) $2x + 7y = 5, \quad -2x + 3y = 15$

(c) $5x + 6y = 28, \quad 9x + 6y = 48$

(d) $-3x + 4y = 10, \quad 3x + 2y = 8$

2. Solve each system by subtracting the equations (the coefficients already match). Check your solution in both original equations.

(a) $2x + 3y = 17, \quad 2x - y = 5$

(b) $3x + 4y = 23, \quad x + 4y = 13$

(c) $6x + y = 16, \quad 6x - 5y = 28$

(d) $-3x + 5y = -17, \quad -3x + 2y = -14$

3. Solve each system by subtracting the equations. Check your solution.

(a) $7x + 2y = 50, \quad 7x - 3y = 30$

(b) $4x + 5y = 17, \quad 9x + 5y = 57$

(c) $-5x + 4y = 34, \quad -5x - 2y = -2$

(d) $3x - 4y = 7, \quad 8x - 4y = 52$

Part B — Eliminate by Addition

4. Solve each system by adding the equations (the coefficients are opposites). Check your solution in both original equations.

(a) $3x + 2y = 19, \quad -3x + 4y = -7$

(b) $2x + 5y = 17, \quad 4x - 5y = 19$

(c) $4x + 3y = 0, \quad -4x + 7y = -40$

(d) $5x + 6y = 20, \quad 3x - 6y = -36$

5. Solve each system by adding the equations. Check your solution.

(a) $6x + 5y = 32, \quad -6x + 2y = -46$

(b) $5x + 3y = 4, \quad 2x - 3y = -32$

(c) $-7x + 4y = -62, \quad 7x + 9y = -3$

Part C — Add or Subtract? You Decide

6. Decide whether to add or subtract, then solve each system. Check your solution.

(a) $3x + 4y = 39, \quad 8x + 4y = 64$

(b) $5x + 2y = 29, \quad -5x + 6y = 27$

(c) $-2x + 7y = 40, \quad -2x + 3y = 24$

(d) $4x + 8y = 12, \quad 5x - 8y = 69$

7. Decide whether to add or subtract, then solve each system. Check your solution.

(a) $9x + 2y = -21, \quad 9x - 5y = 21$

(b) $3x + 7y = 31, \quad 6x - 7y = 41$

(c) $4x - 6y = 64, \quad 7x - 6y = 94$

Part D — Applications

8. The sum of twice a number a and a number b is 23. Twice a minus b is 9.

(a) Write a system of two equations describing this situation.

(b) Solve the system to find a and b.

9. A boat's downstream speed is $b + c$ and its upstream speed is $b - c$, where b is its speed in still water and c is the current's speed. Downstream it travels 22 km/h; upstream, 14 km/h.

(a) Write a system of two equations describing this situation.

(b) Solve the system to find b and c.

Part E — Multiple Choice and Numerical Response

10. *(MC)* Adding $4x + 3y = 10$ and $-4x + 5y = 6$ eliminates a variable and produces the equation
A. $8y = 16$ B. $7x = 16$ C. $2y = 16$ D. $8y = 4$

11. *(MC)* Subtracting the first equation from the second in the system $6x - 2y = 14$, $6x + 3y = 29$ produces the equation
A. $5y = 15$ B. $y = 15$ C. $12x = 15$ D. $5y = 43$

12. *(NR)* If $4x+3y = 29$ and $4x-2y = 14$, then the value of $x + y$ is _____.

13. *(NR)* If $3x + 5y = 6$ and $-3x + 2y = -27$, then the value of $x - y$ is _____.

Part F — Extension and Mixed Review

14. *(Extension)* Consider the system of three equations $x+y = 9$, $y+z = 13$, $x+z = 12$. *Hint:* add all three equations, then divide by 2 to find $x + y + z$; from there subtract each original equation. Find x, y, and z.

15. *(Mixed Review)* Solve each system, deciding whether to add or subtract.
(a) $5x + y = 19$, $8x + y = 25$ (b) $2x + 3y = -7$, $-2x + 7y = 17$

Answer Key. **1.** (a) subtract, x (b) add, x (c) subtract, y (d) add, x. **2.** (a) $(4, 3)$ (b) $(5, 2)$ (c) $(3, -2)$ (d) $(4, -1)$. **3.** (a) $(6, 4)$ (b) $(8, -3)$ (c) $(-2, 6)$ (d) $(9, 5)$. **4.** (a) $(5, 2)$ (b) $(6, 1)$ (c) $(3, -4)$ (d) $(-2, 5)$. **5.** (a) $(7, -2)$ (b) $(-4, 8)$ (c) $(6, -5)$. **6.** (a) $(5, 6)$ (b) $(3, 7)$ (c) $(-6, 4)$ (d) $(9, -3)$. **7.** (a) $(-1, -6)$ (b) $(8, 1)$ (c) $(10, -4)$. **8.** (a) $2a + b = 23$, $2a - b = 9$ (b) $a = 8$, $b = 7$. **9.** (a) $b + c = 22$, $b - c = 14$ (b) $b = 18$ km/h, $c = 4$ km/h. **10.** A. **11.** A. **12.** 8. **13.** 10. **14.** $x = 4$, $y = 5$, $z = 8$. **15.** (a) $(2, 9)$ (b) $(-5, 1)$.

Lesson 6 Assignment *Show your work.*

Part A — Identify the Multiplier

1. State which equation you would multiply, and by what constant, to eliminate the given variable. Do not solve.

(a) $2x + 3y = 10, \quad 5x - y = 4$ (eliminate y)

(b) $x + 4y = 9, \quad 3x - 2y = 1$ (eliminate y)

(c) $4x - y = 7, \quad 2x + 3y = 8$ (eliminate x)

(d) $3x + 2y = 11, \quad x - 5y = -3$ (eliminate x)

Part B — Multiply One Equation

2. Solve each system by multiplying only *one* equation. Check your solution.

(a) $x + 3y = 11, \quad 2x - y = 1$

(b) $2x + y = 13, \quad x - 3y = -4$

(c) $5x + 2y = 24, \quad x - 4y = -4$

(d) $3x - y = 14, \quad 2x + 3y = 13$

3. Solve each system by multiplying only *one* equation. Multiply one equation so a variable's coefficients match, then *subtract* the equations. Watch your signs.

(a) $5x + 2y = 16, \quad x + y = 5$

(b) $3x - 4y = -2, \quad x - 2y = -4$

(c) $2x + 3y = 7, \quad 4x - y = -7$

Part C — Multiply Both Equations

4. Solve each system by multiplying *both* equations. Check your solution.

(a) $2x + 3y = 13, \quad 3x - 2y = 0$

(b) $3x + 2y = 18, \quad 2x - 5y = -7$

(c) $4x + 3y = 24, \quad 5x - 2y = 7$

(d) $5x + 3y = 17, \quad 3x - 4y = 16$

5. Solve each system, choosing whichever variable is easier to eliminate.

(a) $\begin{aligned} 2x + 3y &= 16 \\ 4x - y &= 18 \end{aligned}$

(b) $\begin{aligned} 3x + 5y &= 2 \\ 2x - 3y &= 14 \end{aligned}$

(c) $\begin{aligned} x - 2y &= -2 \\ 3x + 5y &= 38 \end{aligned}$

Part D — Applications

6. A food truck sells tacos and burritos. On Friday, 5 tacos and 2 burritos sold for a total of \$30. On Saturday, 3 tacos and 4 burritos sold for a total of \$32.

(a) Let t be the price of a taco and b the price of a burrito. Write a system of two equations describing the two days' sales.

(b) Solve the system to find the price of a taco and the price of a burrito.

7. A concert sells adult and child tickets. 3 adult tickets and 4 child tickets cost \$54 in total. 5 adult tickets and 3 child tickets cost \$68 in total.

(a) Let a be the price of an adult ticket and c the price of a child ticket. Write a system of two equations describing the two ticket totals.

(b) Solve the system to find the price of an adult ticket and the price of a child ticket.

8. A garden bed has length l and width w, both in metres. Twice the length plus three times the width is 57 m. Three times the length minus twice the width is 27 m.

(a) Write a system of two equations describing the garden bed.

(b) Solve the system to find the length and width of the garden bed.

Part E — Multiple Choice and Numerical Response

9. *(MC)* Which pair of multipliers sets up *opposite* y-coefficients so that $2x + 5y = 19$ and $3x - 2y = 1$ can be eliminated by adding?
A. Eq. 1 by 2, Eq. 2 by 5 B. Eq. 1 by 5, Eq. 2 by 2
C. Eq. 1 by 3, Eq. 2 by 4 D. Eq. 1 by 2, Eq. 2 by 3

10. *(MC)* When y is eliminated from $2x + 3y = 13$ and $x - 6y = -16$ by multiplying the first equation by 2 and then adding, the result is
A. $5x = 10$ B. $3x = 10$ C. $5x = 42$ D. $x = 10$

11. *(NR)* If $3x+2y = 14$ and $4x-3y = 13$, then the value of $x + y$ is _____.

12. *(NR)* If $5x - 2y = 8$ and $2x + 3y = 26$, then the value of $2x - y$ is _____.

Part F — Extension and Mixed Review

13. *(Extension)* Solve the system $x + y + z = 12$, $2x - y + z = 7$, $x + 2y - z = 6$ by elimination. *Hint:* add pairs of equations to eliminate z or y first, then solve the resulting two-equation system.

14. *(Mixed Review)* Solve each system, choosing whichever multiplying strategy (one equation or both) is most efficient.
(a) $2x + y = 13$, $5x - 3y = 5$ (b) $4x + 3y = -1$, $3x - 5y = 21$

Answer Key. 1. (a) multiply Eq. 2 by 3, add (b) Eq. 2 by 2, add (c) Eq. 2 by 2, then subtract (d) Eq. 2 by 3, then subtract. **2.** (a) $(2, 3)$ (b) $(5, 3)$ (c) $(4, 2)$ (d) $(5, 1)$. **3.** (a) $(2, 3)$ (b) $(6, 5)$ (c) $(-1, 3)$. **4.** (a) $(2, 3)$ (b) $(4, 3)$ (c) $(3, 4)$ (d) $(4, -1)$. **5.** (a) $(5, 2)$ (b) $(4, -2)$ (c) $(6, 4)$. **6.** (a) $5t + 2b = 30$, $3t + 4b = 32$ (b) taco \$4, burrito \$5. **7.** (a) $3a + 4c = 54$, $5a + 3c = 68$ (b) adult \$10, child \$6. **8.** (a) $2l + 3w = 57$, $3l - 2w = 27$ (b) $l = 15$ m, $w = 9$ m. **9.** A. **10.** A. **11.** $x = 4$, $y = 1$, so $x + y = 5$. **12.** $x = 4$, $y = 6$, so $2x - y = 2$. **13.** $x = 3$, $y = 4$, $z = 5$. **14.** (a) $(4, 5)$ (b) $(2, -3)$.

Lesson 7 Assignment *Show your work.*

Part A — Translating Only

1. For each situation below, assign variables and write a system of two equations that models it. **Do not solve the system.**
(a) The length of a garden path is twice its width. The sum of the length and width is 21 m.

(b) A jar contains dimes and quarters only. There are 40 coins in total, worth \$7.75 (775 cents).

Part B — Number and Age Problems

2. The sum of two numbers is 45. The larger number is 9 more than the smaller number. Find the two numbers.

Answer: _____

3. One number is 3 times another number. Their sum is 48. Find the two numbers.

Answer: _____

4. Sofia is 7 years older than her brother Diego. The sum of their ages is 33. Find both ages.

Answer: _____

5. A father is 3 times as old as his daughter. The sum of their ages is 52. Find both ages.

Answer: _____

Part C — Cost and Purchase Problems

6. Movie tickets cost \$11 for adults and \$7 for children. A family bought 9 tickets in total and paid \$79. How many adult tickets and how many child tickets did they buy?

Answer: _____

7. A bakery sells muffins for \$3 and cookies for \$2. One morning, the bakery sold a total of 50 items for \$126. How many muffins and how many cookies were sold?

Answer: _____

8. Concert tickets are sold as floor seats (\$45) and balcony seats (\$30). A total of 120 tickets were sold, generating \$4650 in revenue. How many floor seats and how many balcony seats were sold?

Answer: _____

Part D — Mixture Problems

9. A trail mix combines raisins (\$4/kg) with almonds (\$9/kg). The mix weighs 25 kg in total and sells for \$170. How many kg of each ingredient are in the mix?

Answer: _____

10. A coffee shop blends dark roast beans (\$15/kg) with light roast beans (\$9/kg) to make a house blend. The blend weighs 12 kg in total and costs \$138. How many kg of each type of bean are in the blend?

Answer: _____

Part E — Multiple Choice and Numerical Response

11. *(Multiple Choice)* A word problem states: *"The sum of two numbers is* 40*. The larger number is* 4 *less than three times the smaller number."* If x is the larger number and y is the smaller number, which system correctly represents this situation?
A. $x + y = 40,\ x = 3y + 4$ B. $x - y = 40,\ x = 3y - 4$ C. $x + y = 40,\ x = 3y - 4$ D. $x + y = 40,\ y = 3x - 4$

12. *(Numerical Response)* The sum of two numbers is 52. The larger number is 10 more than the smaller number. The value of the smaller number is _____.
(Record your answer from left to right.)

13. *(Numerical Response)* At a bake sale, cupcakes sell for \$2 and cookies sell for \$1. A total of 45 items were sold for \$76. The number of cupcakes sold is _____.
(Record your answer from left to right.)

Part F — Mixed Review

14. *(Mixed Review)* A school ordered a total of 200 tickets for the spring play: adult tickets at \$12 and student tickets at \$7, for total ticket revenue of \$2000.
(a) Write a system of two equations representing this situation.
(b) Solve the system to find the number of adult tickets and the number of student tickets ordered.

Answer: _____

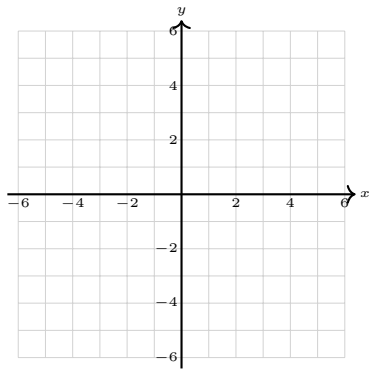
Answer Key. **1.** (a) $l = 2w$, $l + w = 21$ (b) $d + q = 40$, $10d + 25q = 775$ (in cents). **2.** 27 and 18. **3.** 36 and 12. **4.** Sofia 20, Diego 13. **5.** Father 39, daughter 13. **6.** 4 adult, 5 child. **7.** 26 muffins, 24 cookies. **8.** 70 floor, 50 balcony. **9.** 11 kg raisins, 14 kg almonds. **10.** 5 kg dark roast, 7 kg light roast. **11.** C. **12.** 21. **13.** 31 cupcakes. **14.** (a) $a + s = 200$, $12a + 7s = 2000$ (b) 120 adult, 80 student.

Lesson 8 Assignment *Show your work.*

Part A — Graphing and Classifying

1. The lines $y = -x + 6$ and $y = 3x - 2$ cross at integer coordinates. Solve the system algebraically (set the two expressions for y equal) to confirm the point of intersection, and state the solution as an ordered pair.

2. Graph $y = 2x - 4$ and $y = -x + 5$ without technology, and state the solution to the system.



3–6. Without graphing, state whether each system has one solution, no solution, or infinitely many solutions.

3. $y = 5x - 2$, $y = 5x + 4$

4. $y = -2x + 9$, $y = 4x - 3$

5. $y = 7x - 10$, $y = 7x - 10$

6. $y = -3x + 2$, $y = 3x + 2$

7. Consider the system $2x + 2y = 10$, $y = -x + 5$. Rewrite the first equation in slope-intercept form, then state whether the system has one solution, no solution, or infinitely many solutions.

Part B — Substitution

8–9. Solve using substitution.

8. $y = 5x - 6$, $3x + y = 10$

9. $x = 2y + 7$, $x - y = 10$

10–11. Solve using substitution, isolating a variable first.

10. $x + 4y = 18$, $3x - y = 15$

11. $2x + y = 17$, $x - 3y = -16$

Part C — Elimination

12–13. Solve using elimination.

12. $2x + 3y = 18$, $4x - 3y = 18$

13. $5x + 2y = 7$, $5x - 3y = 27$

14–15. Solve using elimination, multiplying each equation by a suitable constant first.

14. $3x + 2y = 16$, $2x + 5y = 18$

15. $4x + 3y = 7$, $5x - 2y = -20$

Part D — Word Problems

16. The sum of two numbers is 50. The larger number is 2 more than three times the smaller number. Let L be the larger number and s the smaller number.
(a) Write a system of two equations describing this situation.
(b) Solve the system to find both numbers.

17. A school play sold 210 tickets, some at the general price of \$8 and some at the student price of \$5, for total sales of \$1410.
(a) Let g be the number of general tickets and t the number of student tickets. Write a system of two equations.
(b) Solve to find how many of each type were sold.

18. A rectangular banner has a perimeter of 44 cm. Its length is 4 cm more than its width.
(a) Write a system of two equations describing the banner.
(b) Solve the system to find the length and width of the banner.

19. Consider the system $4x - y = 13$, $2x + 3y = 17$. Solve using whichever method you think is fastest, and explain in one sentence why you chose that method.

Part E — Mixed Review

20. Gym A charges a \$20 sign-up fee plus \$35 per month; Gym B charges a \$50 sign-up fee plus \$35 per month.
(a) Write an equation for the total cost at each gym in terms of the number of months m.
(b) Without solving, classify the system and explain what this means about the two gyms' pricing.

21. A community garden is rectangular. Its perimeter is 78 m, and its length is 3 m less than twice its width. Find the length and width of the garden.

22. If $3x + 4y = 23$ and $x - 2y = 1$, determine the value of $x + y$.

23. If $5x - 2y = 11$ and $2x + 3y = 12$, determine the value of $2x - y$.

24. A vending machine sold 90 bags of snacks: small bags at \$2 and large bags at \$3. Total sales were \$249. How many of each were sold?

Answer Key. **1.** $(2, 4)$. **2.** $(3, 2)$. **3.** No solution (same slope 5, different intercepts). **4.** One solution (slopes -2 and 4). **5.** Infinitely many (identical equations). **6.** One solution (slopes -3 and 3). **7.** $y = -x + 5$ — same slope and intercept as the second equation, so infinitely many solutions. **8.** $(2, 4)$. **9.** $(13, 3)$. **10.** $(6, 3)$. **11.** $(5, 7)$. **12.** $(6, 2)$. **13.** $(3, -4)$. **14.** $(4, 2)$. **15.** $(-2, 5)$. **16.** (a) $L + s = 50$, $L = 3s + 2$ (b) larger = 38, smaller = 12. **17.** (a) $g + t = 210$, $8g + 5t = 1410$ (b) 120 general, 90 student. **18.** (a) $2l + 2w = 44$, $l = w + 4$ (b) length = 13 cm, width = 9 cm. **19.** $(4, 3)$; y has coefficient -1 in the first equation, so isolating it makes substitution quick. **20.** (a) $C_A = 35m + 20$, $C_B = 35m + 50$ (b) same slope (35), different intercepts $\Rightarrow$ no solution — the gyms never cost the same; Gym B is always \$30 more. **21.** Length = 25 m, width = 14 m. **22.** $x + y = 7$ (from $x = 5$, $y = 2$). **23.** $2x - y = 4$ (from $x = 3$, $y = 2$). **24.** 21 small bags, 69 large bags.